

Hypothalamus – Neuroanatomy, Neuroendocrine and Regional Function

Vladimir Grubisic, MD PhD
Biomedical Sciences
vladimir.grubisic@nyit.edu

Do.
Make.
Heal.
Innovate.

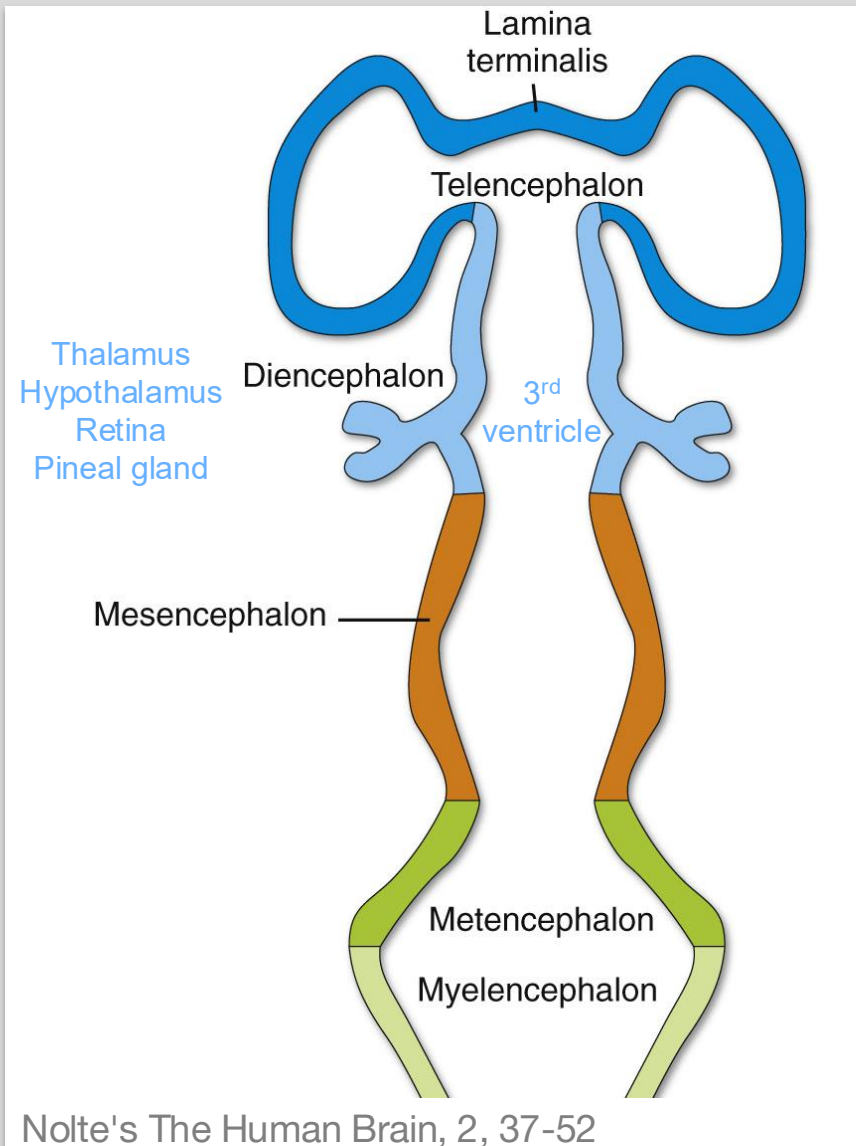
Reinvent the Future.

Session Objectives

At the end of this lecture, students will be able to:

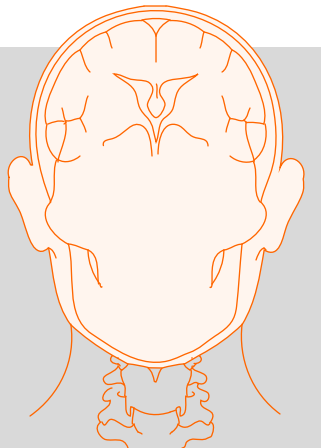
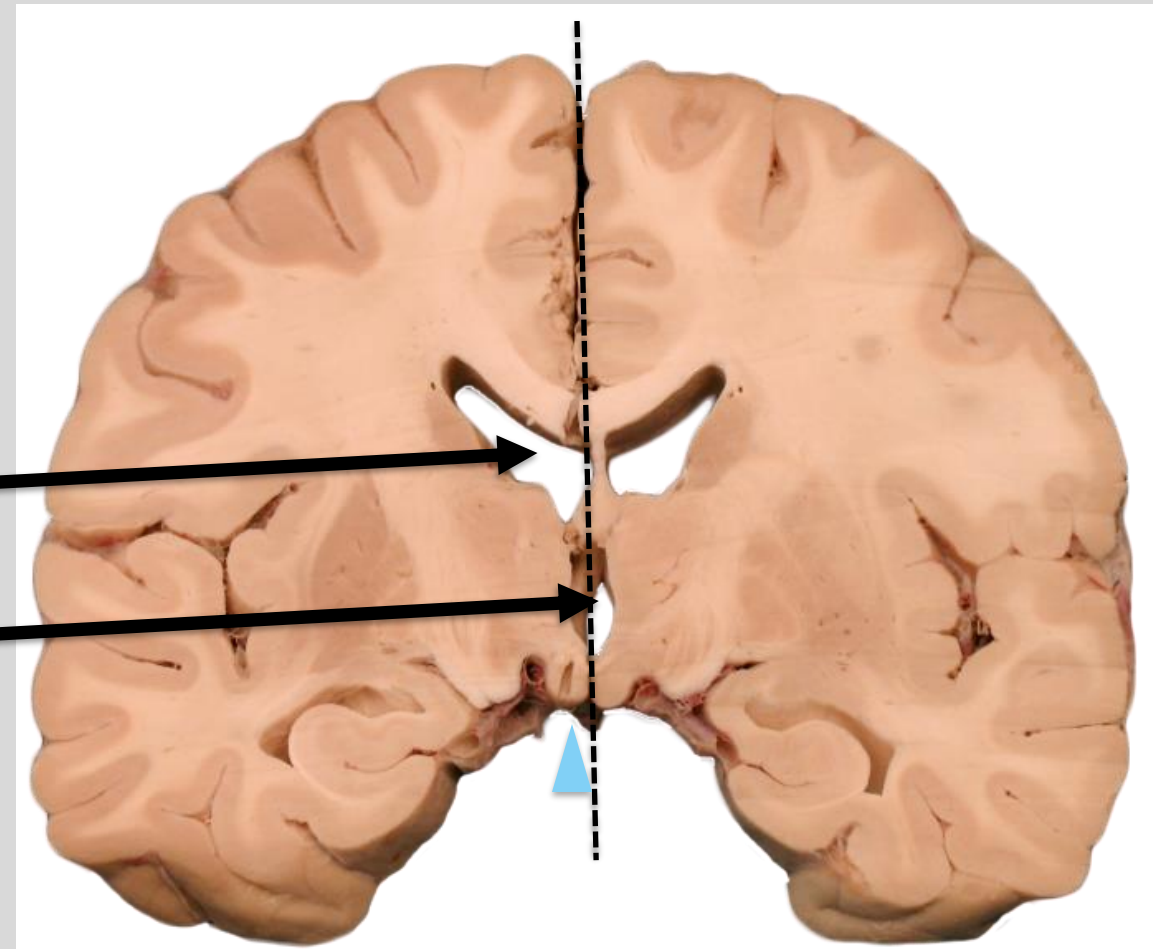
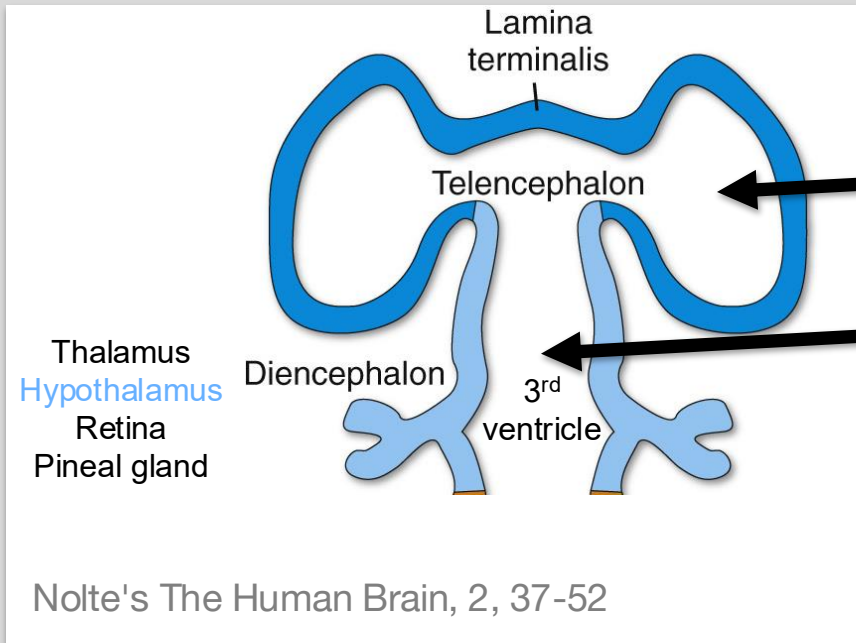
1. Describe the gross anatomy of the hypothalamus, including major hypothalamic nuclei.
2. Describe the functions of the hypothalamus, including hypothalamic hormone secretion and its regulation of pituitary hormones.
3. Understand clinical correlations between the location of hypothalamic nuclei and their dysfunction.
4. Describe the anatomy and regulation of feeding behavior and clinical correlation to obesity.

Development

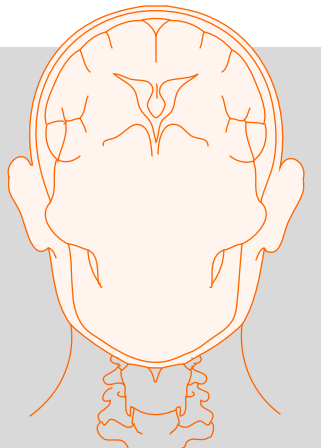
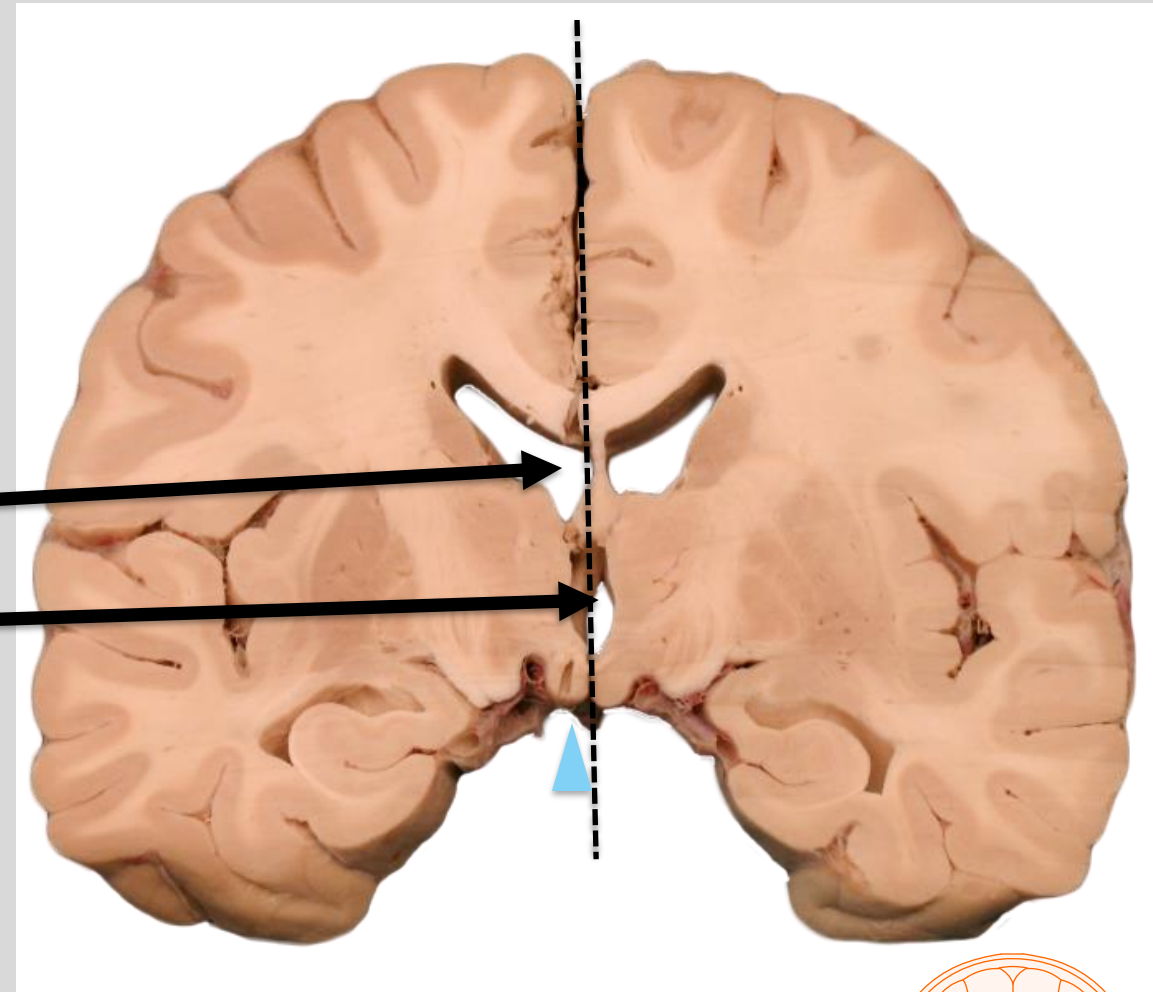
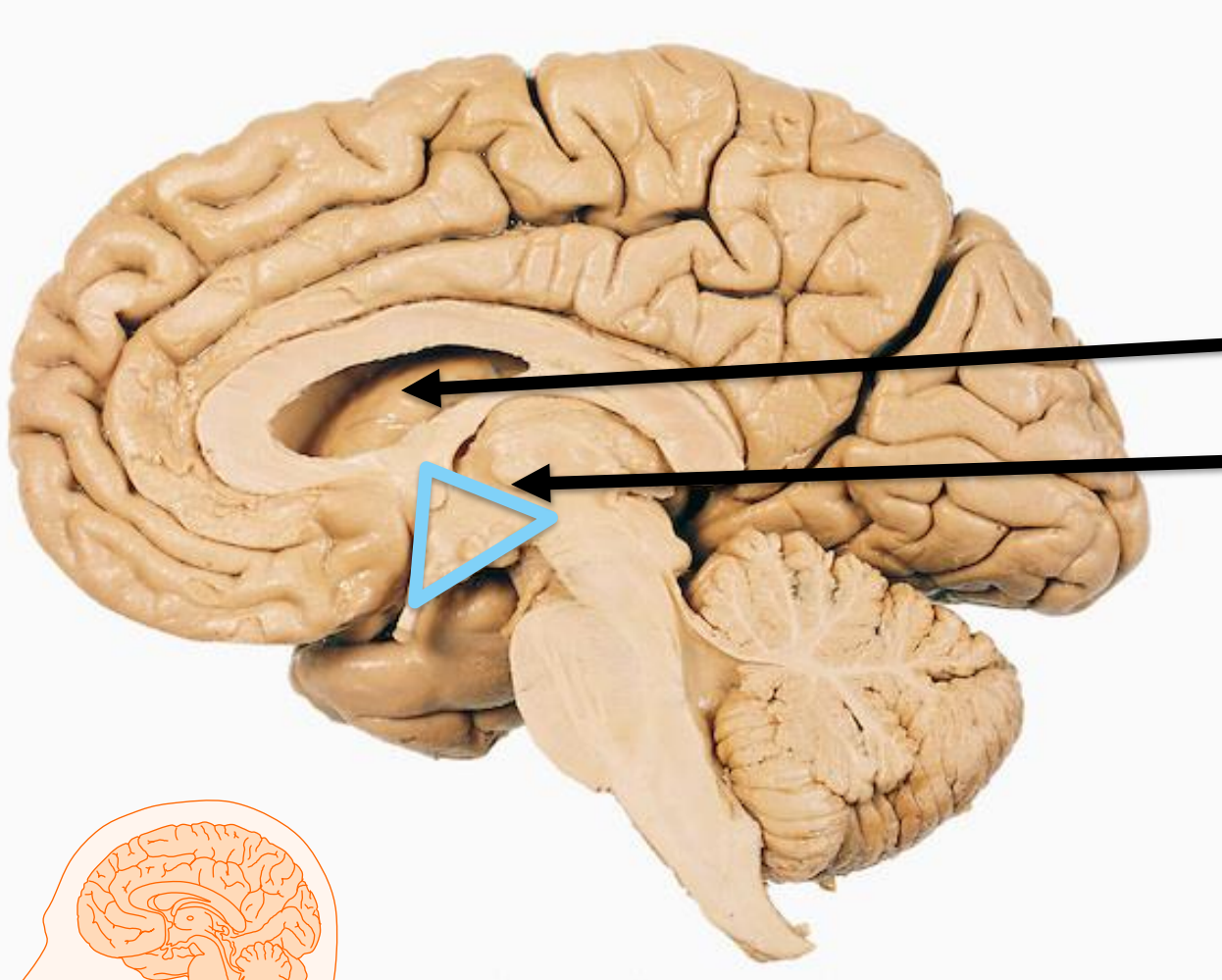


Neuroanatomy

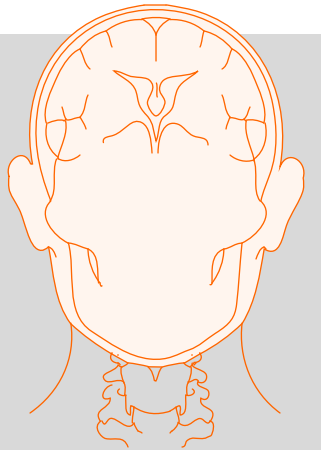
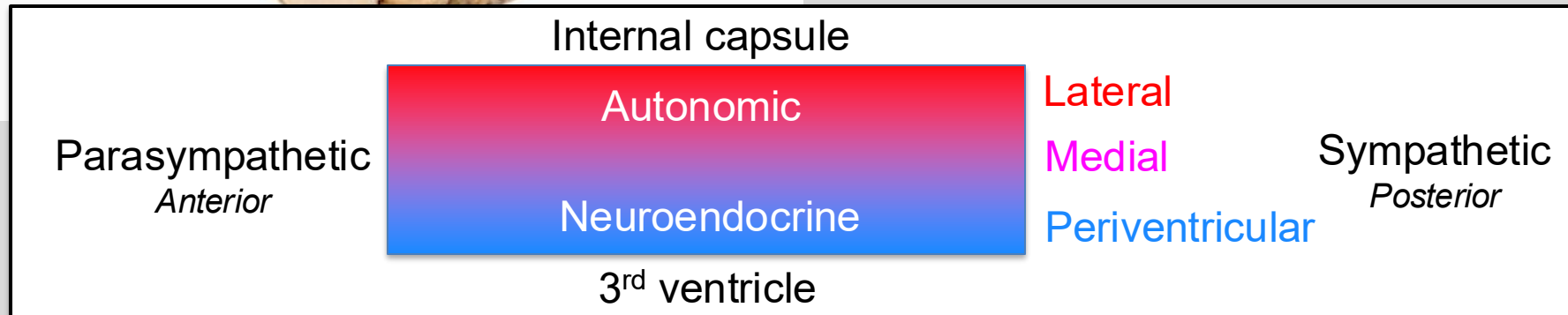
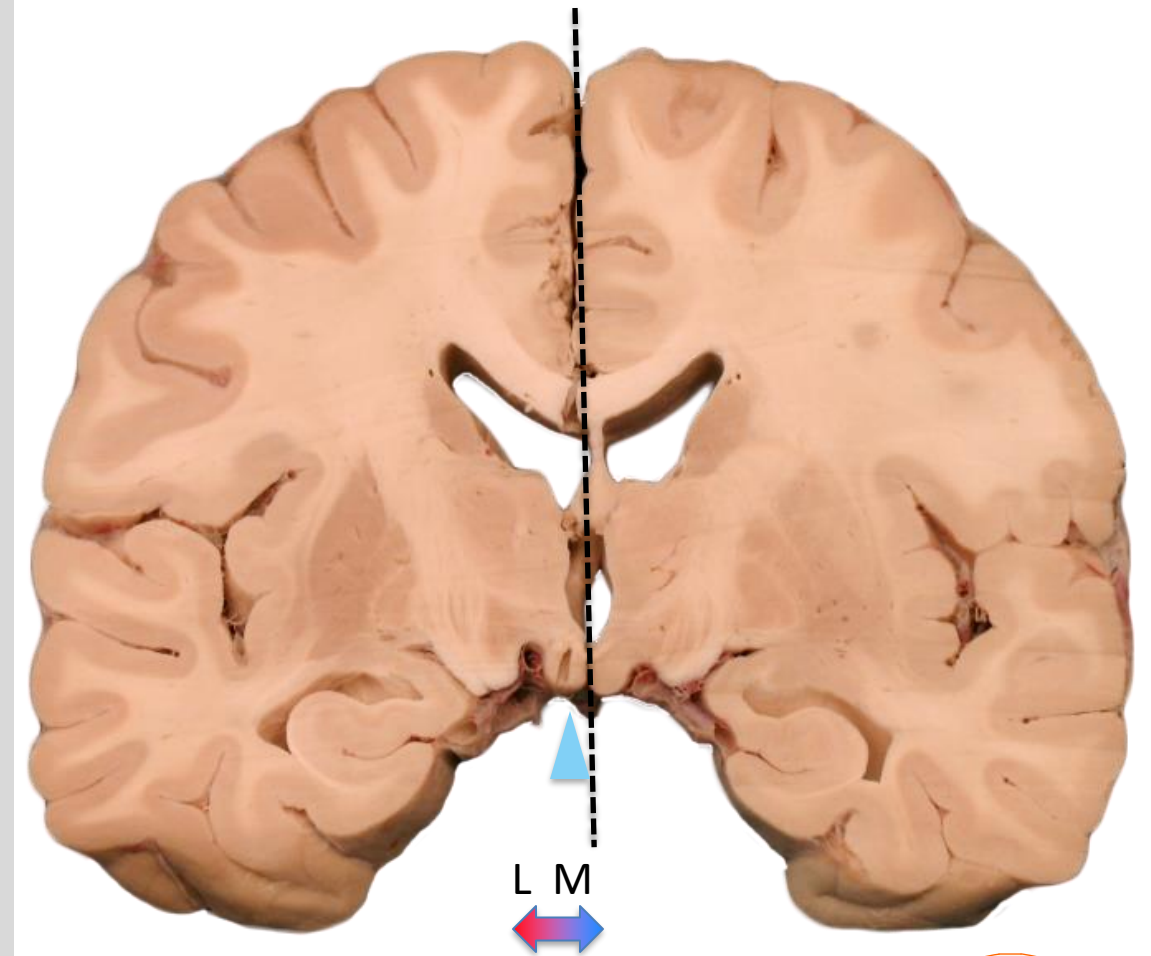
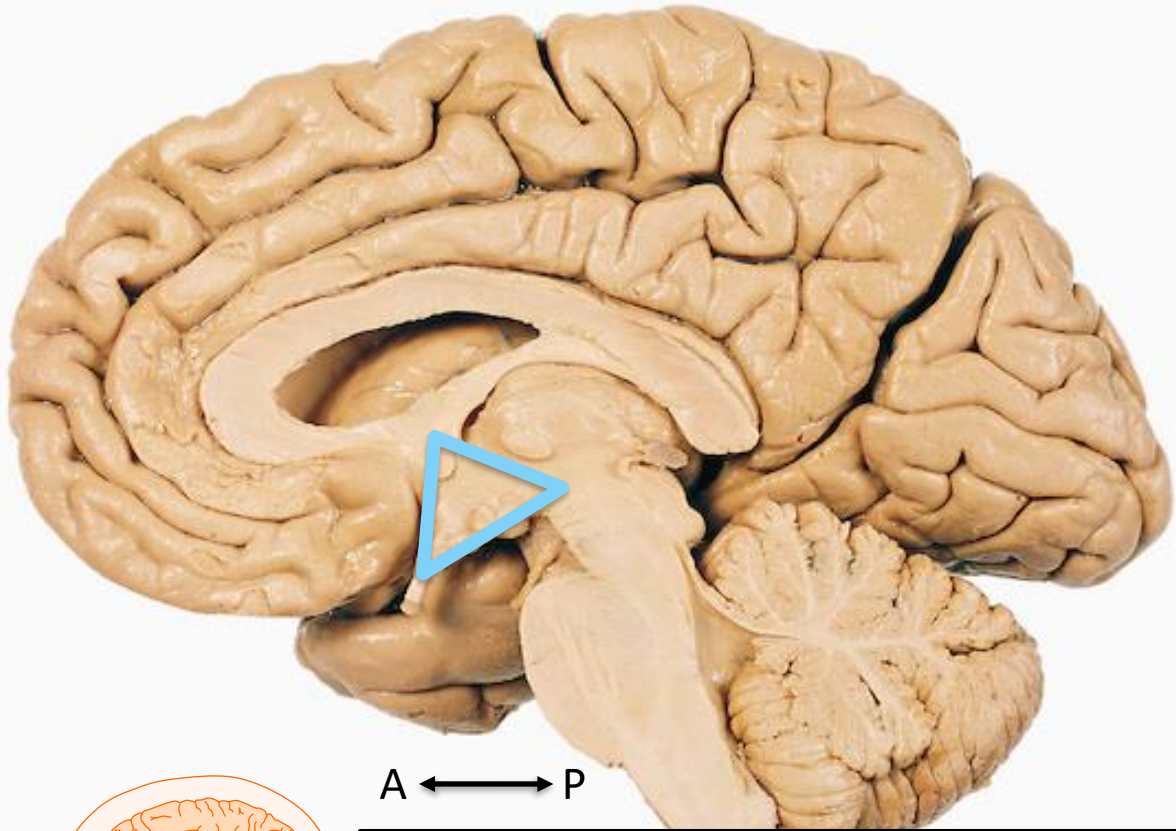
Development



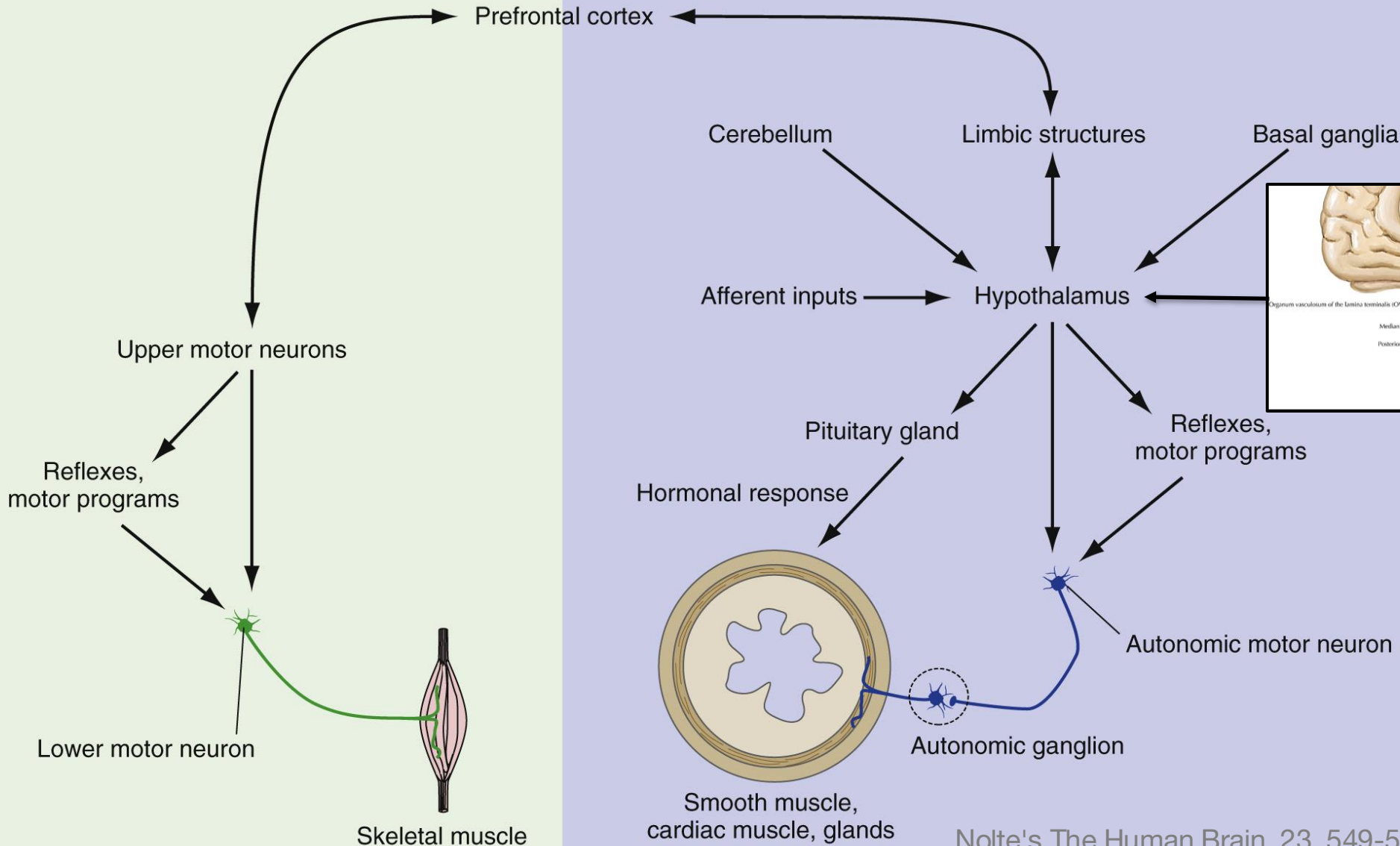
Neuroanatomy



Neuroanatomy



Hypothalamus - center that maintains homeostasis



Circumventricular organs:

- Posterior pituitary
- Median eminence
- Organum vasculosum of the lamina terminalis (OVLT)
- Pineal gland
- Subfornical organ

Major Hypothalamus Functions

Thirst and water balance

Adrenohypophysis control

Neurohypophysis releases hormones synthesized by hypothalamus

Hunger

Autonomic regulation

Visceral functions & affective behaviors

Blood pressure

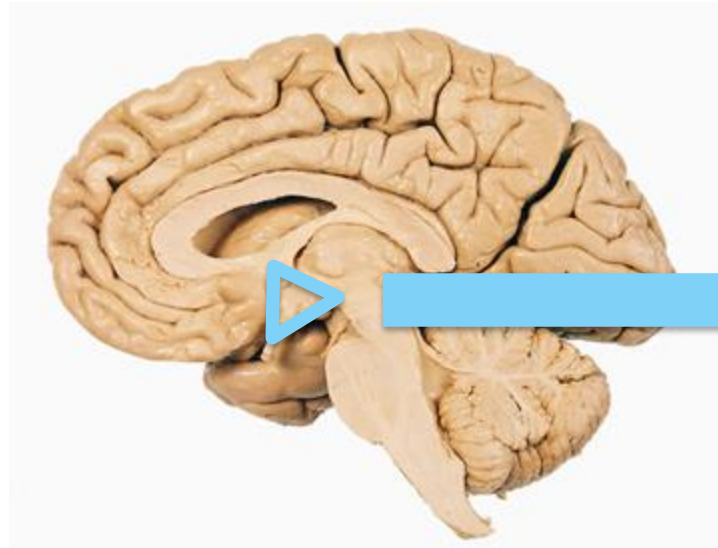
Temperature regulation

Sleep-wake mechanisms

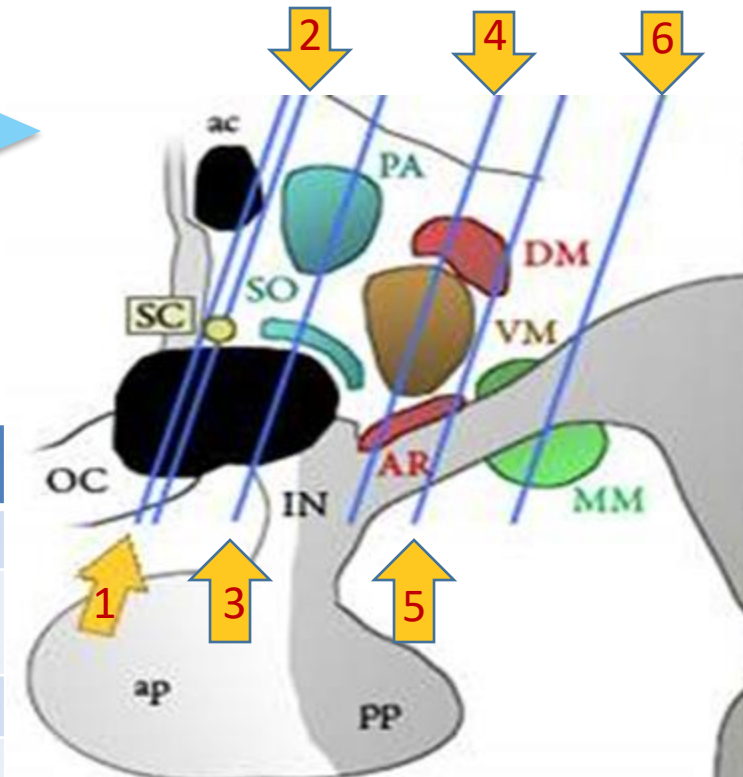
Sexual urges, **S**leep

Circadian rhythms

Major Hypothalamic Nuclei - Neuroanatomy



Six Sections



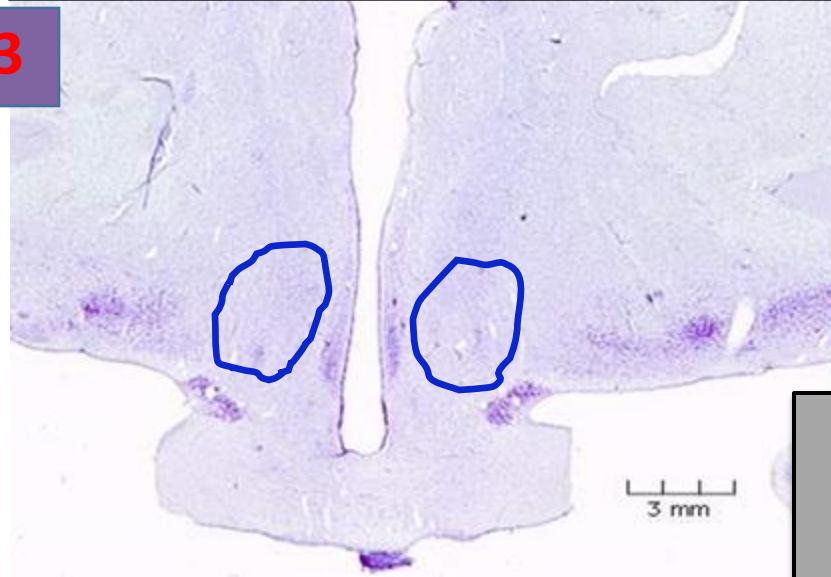
KEY

OC	Optic Chiasm	PA	Paraventricular Nucleus
SC	Suprachiasmatic Nucleus	AR	Arcuate Nucleus
SO	Supraoptic Nucleus	DM	Dorsomedial Nucleus
IN	Infundibulum	VM	Ventromedial Nucleus
ap	Anterior Pituitary	MM	Medial Mammillary Nucleus
pp	Posterior Pituitary	ac	Anterior Commissure

Major Hypothalamic Nuclei - Temperature

Anterior Hypothalamic Nucleus

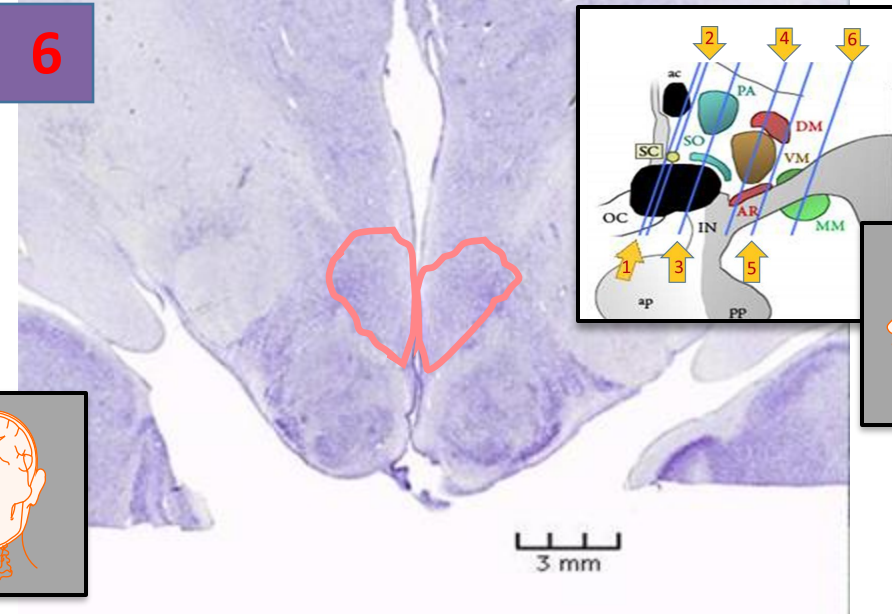
3



- Plays a role in temperature regulation through cooling
- Lesions to this nucleus results in **hyperthermia**

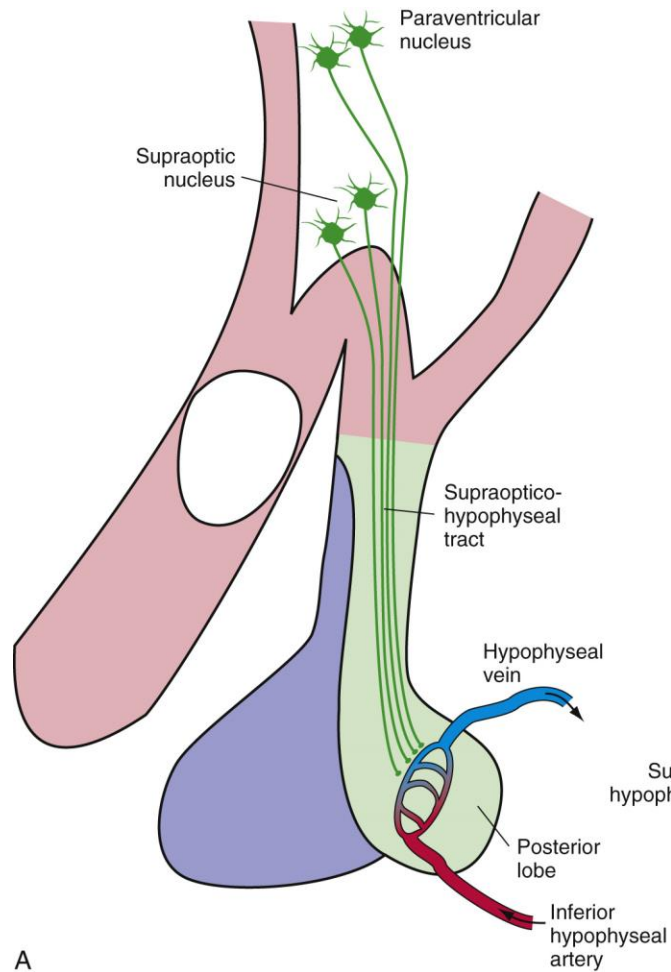
Posterior Hypothalamic Nucleus

6

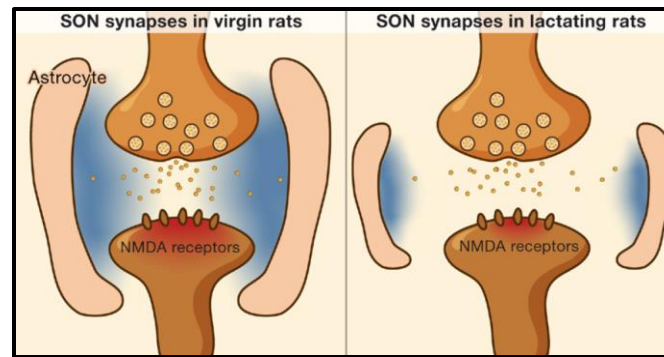
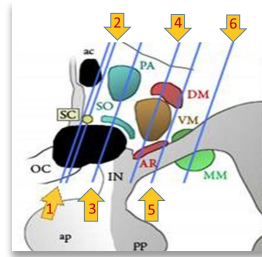
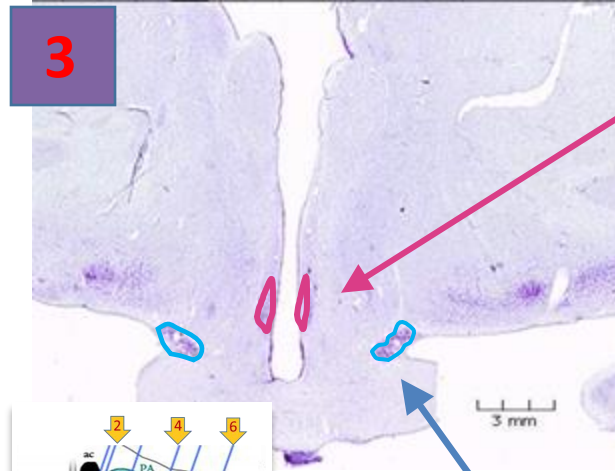


- Plays a role in thermal regulation (i.e., conservation and increased production of heat)
- Lesions to this nucleus results in the inability to thermo-regulate; **hypothermia**

Major Hypothalamic Nuclei – Neurohypophysis



A



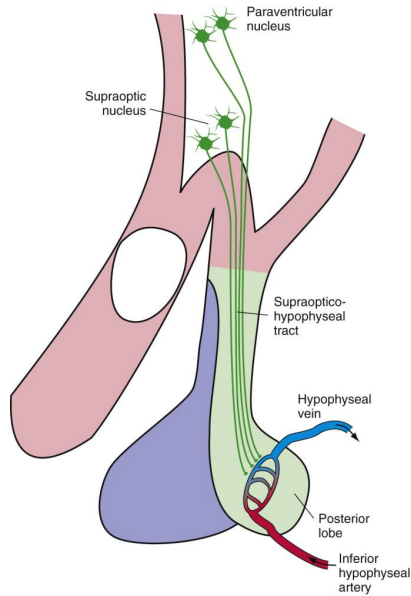
Paraventricular Nucleus

- Projects directly to the autonomic nuclei of the brain, brainstem and all levels of the spinal cord
- Synthesizes:
 - Corticotropin-Releasing Factor (**CRF**) and Thyrotropin-releasing hormone (**TRH**)
 - Anti-Diuretic Hormone (**ADH**) or Arginine VasoPression (**AVP**), and **Oxytocin**

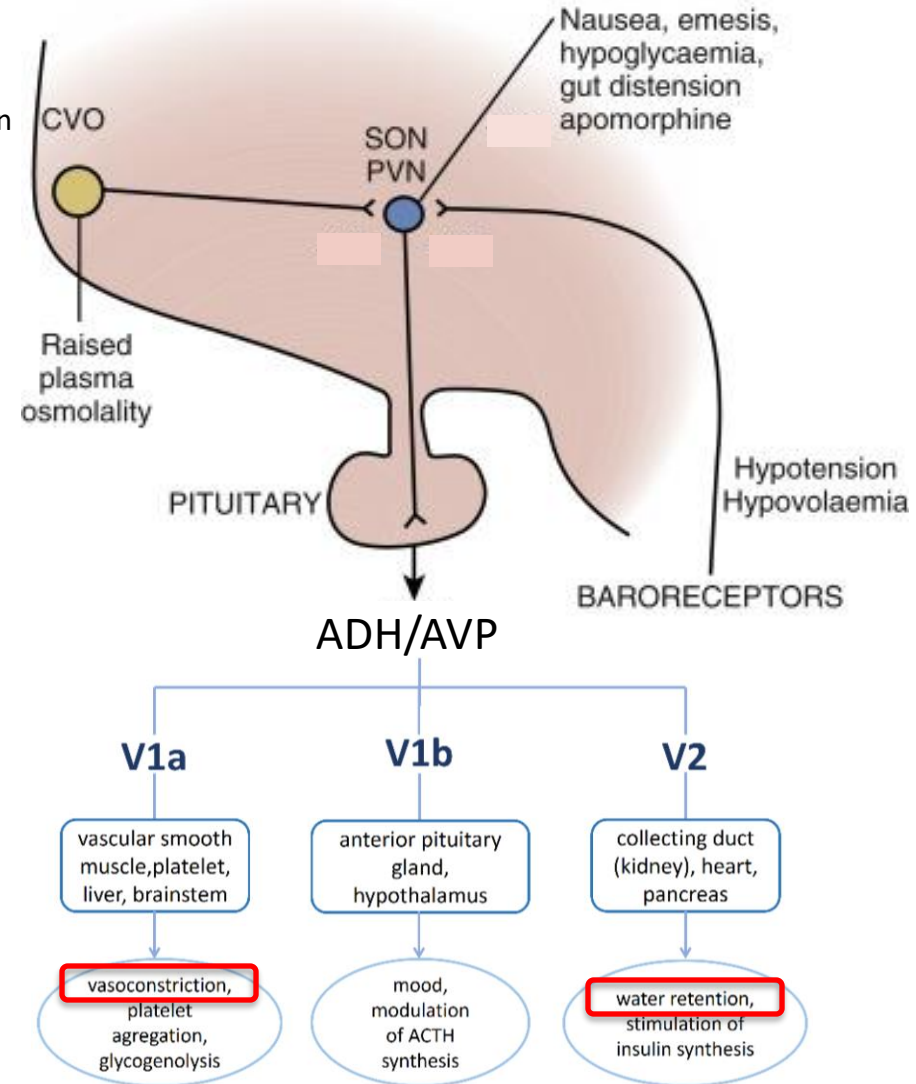
Supraoptic Nucleus

- Synthesizes:
 - ADH/AVP and Oxytocin

ADH/AVP and water homeostasis



CVO, Circumventricular organ
(site of osmoreceptors)



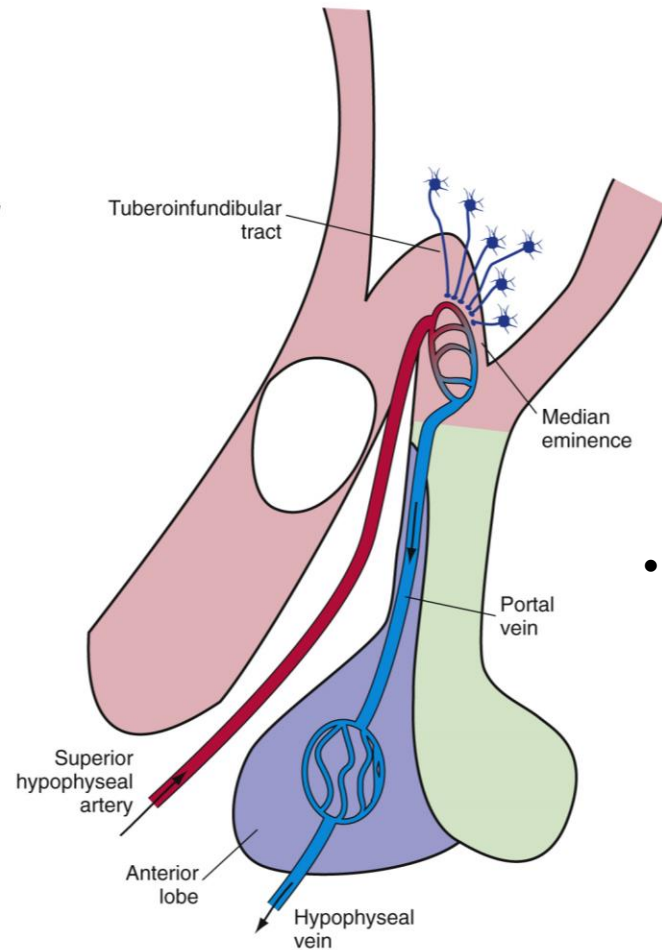
ADH/AVP -> Osmoregulation

↓ Neurogenic diabetes insipidus

↑ Syndrome of Inappropriate

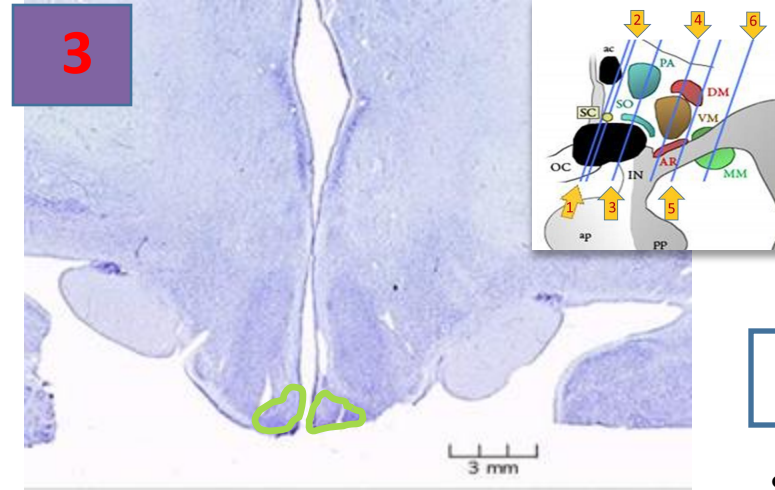
ADH secretion (SIADH)

Major Hypothalamic Nuclei – Adenohypophysis



Nolte's The Human Brain, 23, 549-572

Arcuate Nucleus



- Contains neurons that produce factors that stimulate or inhibit the hypothalamus

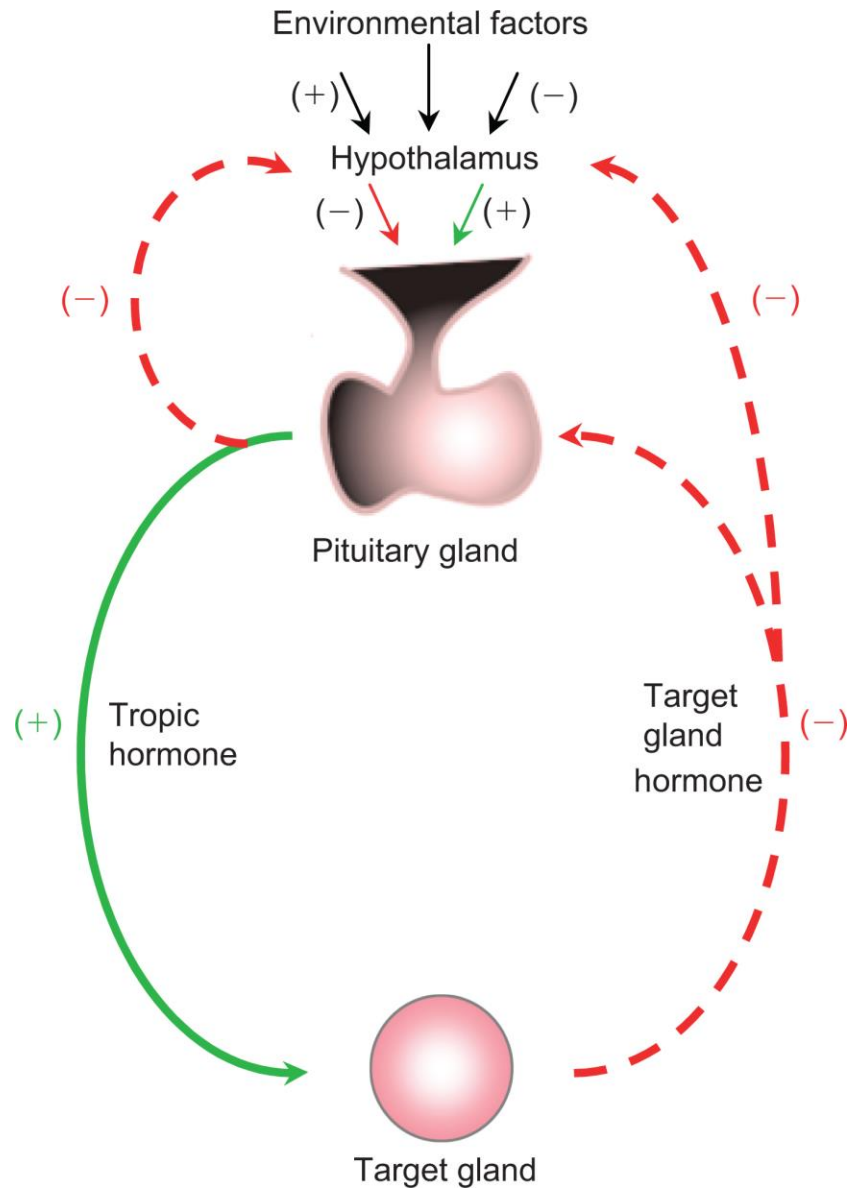
Examples:

- Neurons produce dopamine which inhibits the secretion of prolactin in the anterior pituitary
- Neurons produce growth hormone releasing hormone (GHRH)

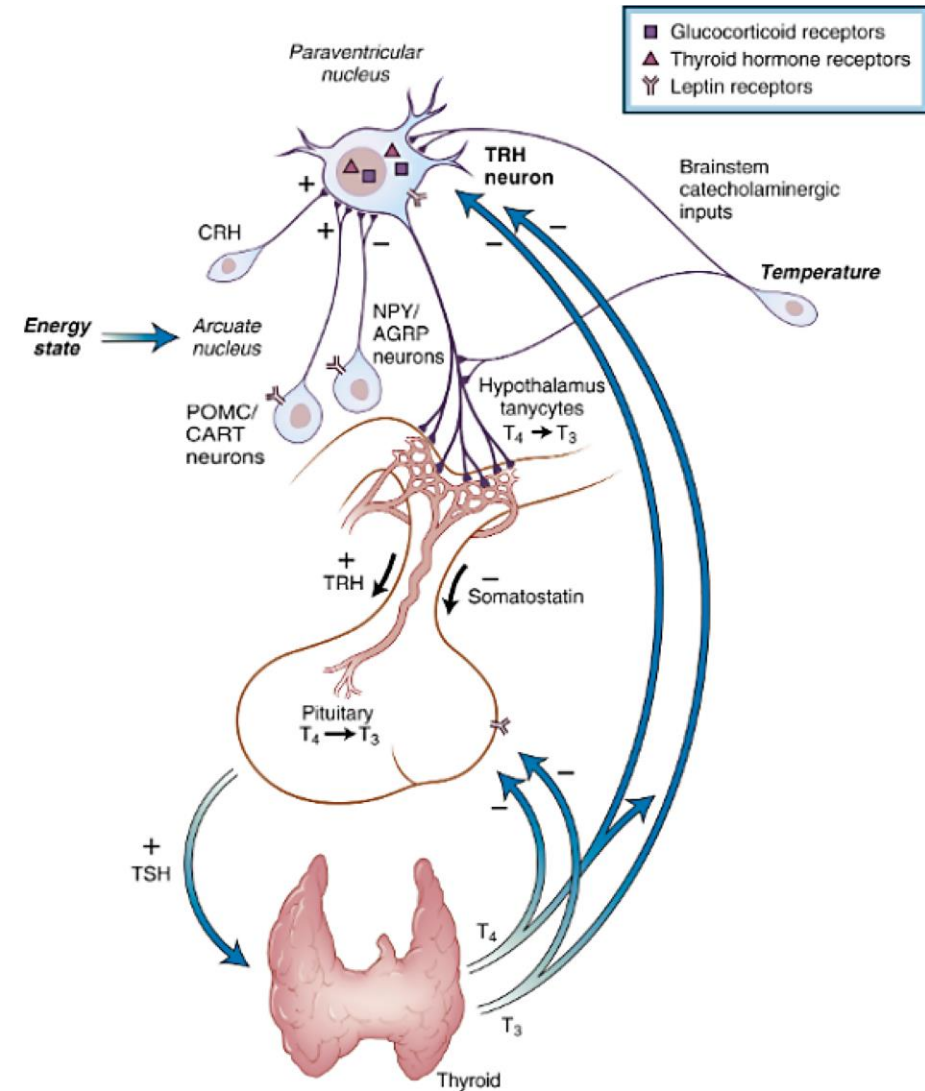
Preoptic Area

- Production of gonadotropin-releasing hormone (GnRH)
- Sexual behavior
- Receptive area:
 - Drinking behavior
 - Temperature regulation

Regulation of anterior pituitary hormone secretion

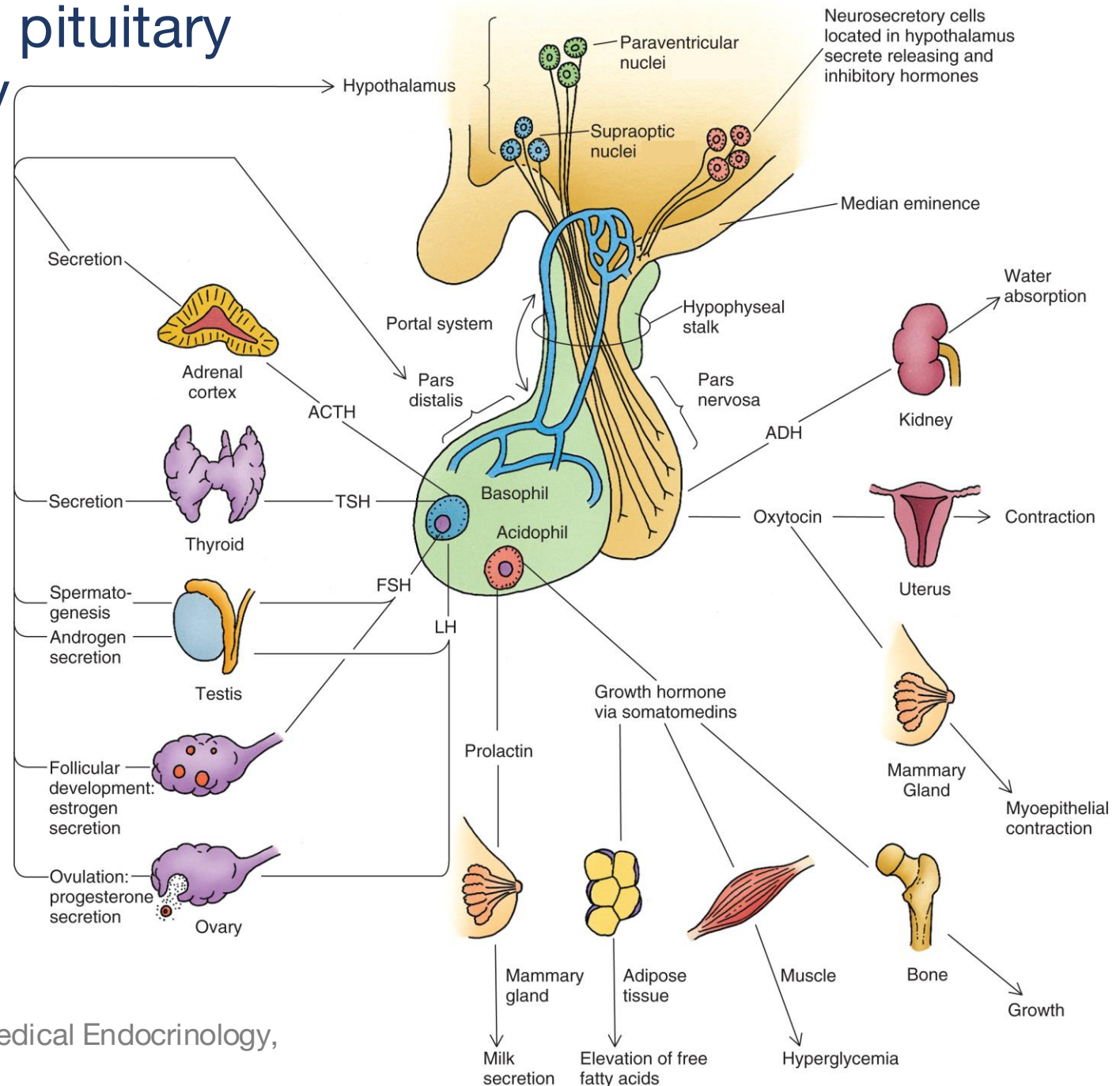


The hypothalamus–pituitary–thyroid axis



Hypothalamus and pituitary

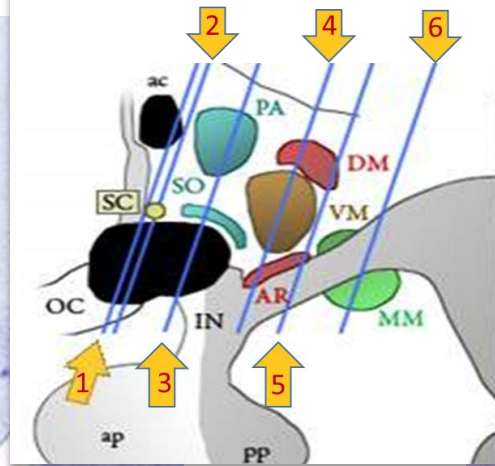
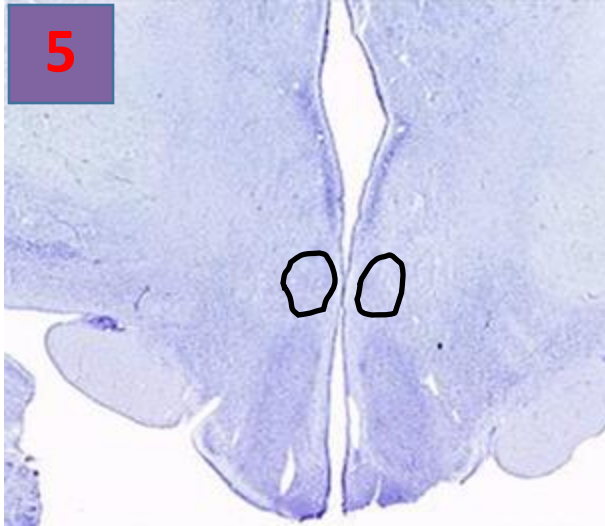
- summary



Modified from Goodman's Basic Medical Endocrinology,
Chapter 2, 43-73

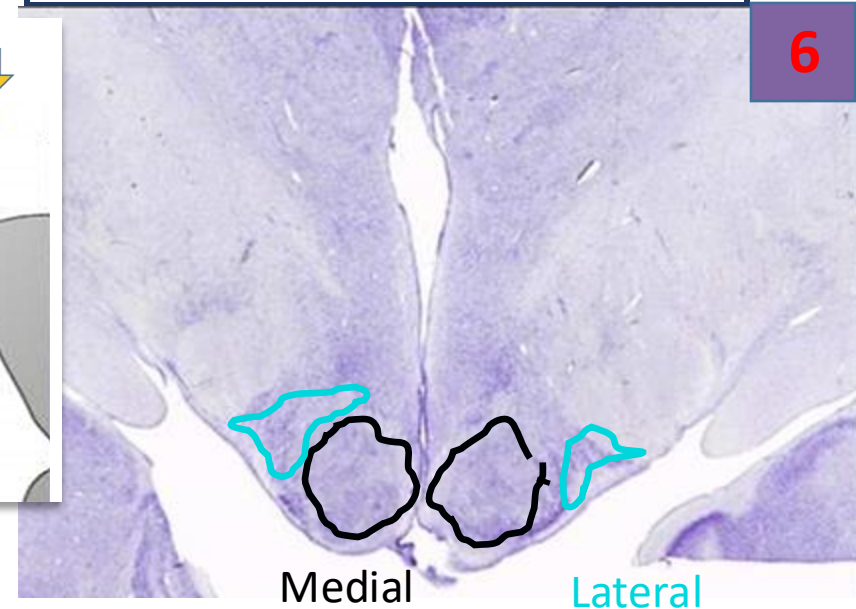
Major Hypothalamic Nuclei – Behavior

Dorsomedial Nucleus



- Stimulation in animals results in aggressive behavior
- Plays a role in reproduction, feeding behavior, and the stress response to fear

Mammillary Nuclei

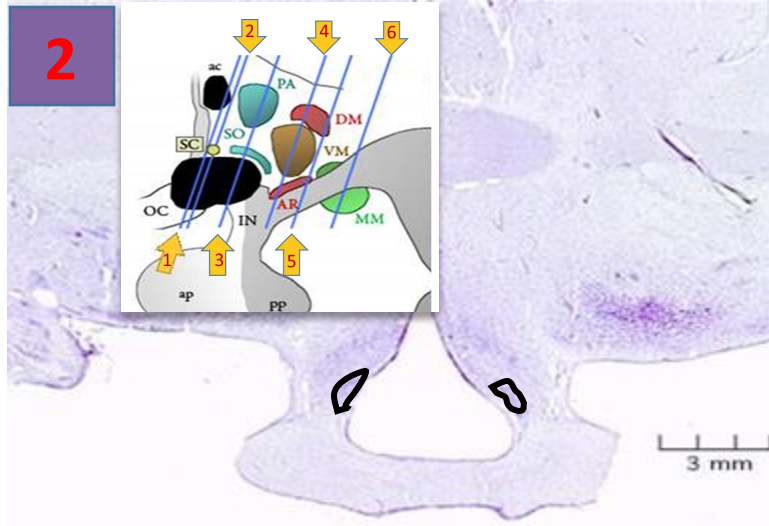


- Receives input from the hippocampus via the fornix
- Projects to the anterior nucleus of the thalamus through the mammillothalamic tract
- Important for episodic memory

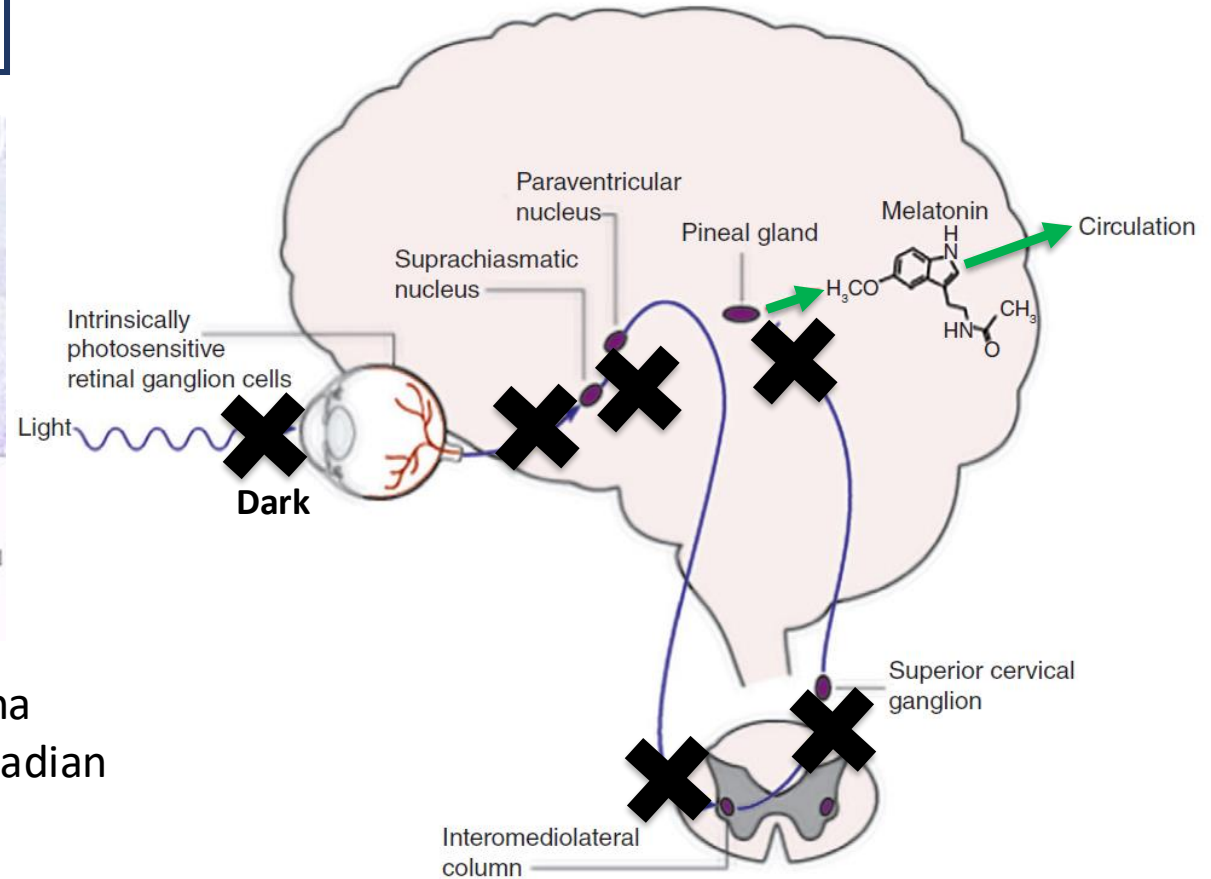
Clinical Correlation: Patients with Wernicke-Korsakoff Syndrome, resulting from a thiamine (vitamin B₁) deficiency, have lesions in the mammillary nucleus. Lesions result in amnesia.

Major Hypothalamic Nuclei – Circadian rhythm

Suprachiasmatic Nucleus

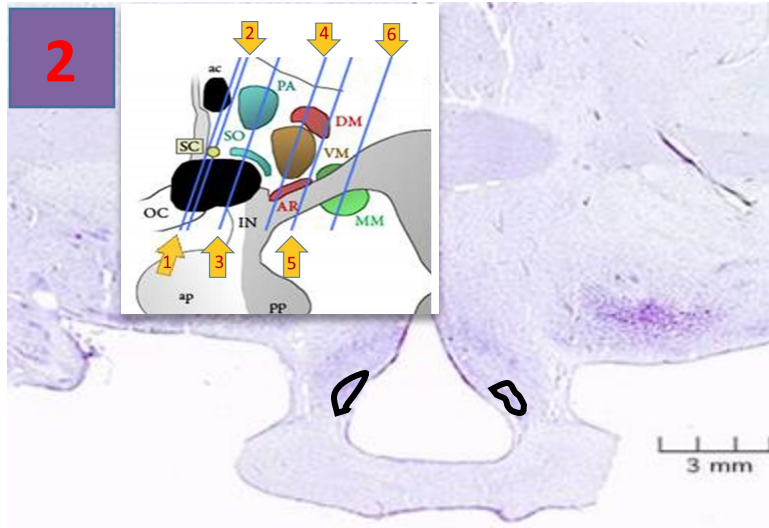


- Receives direct input from the retina
- Plays a role in the regulation of circadian rhythms

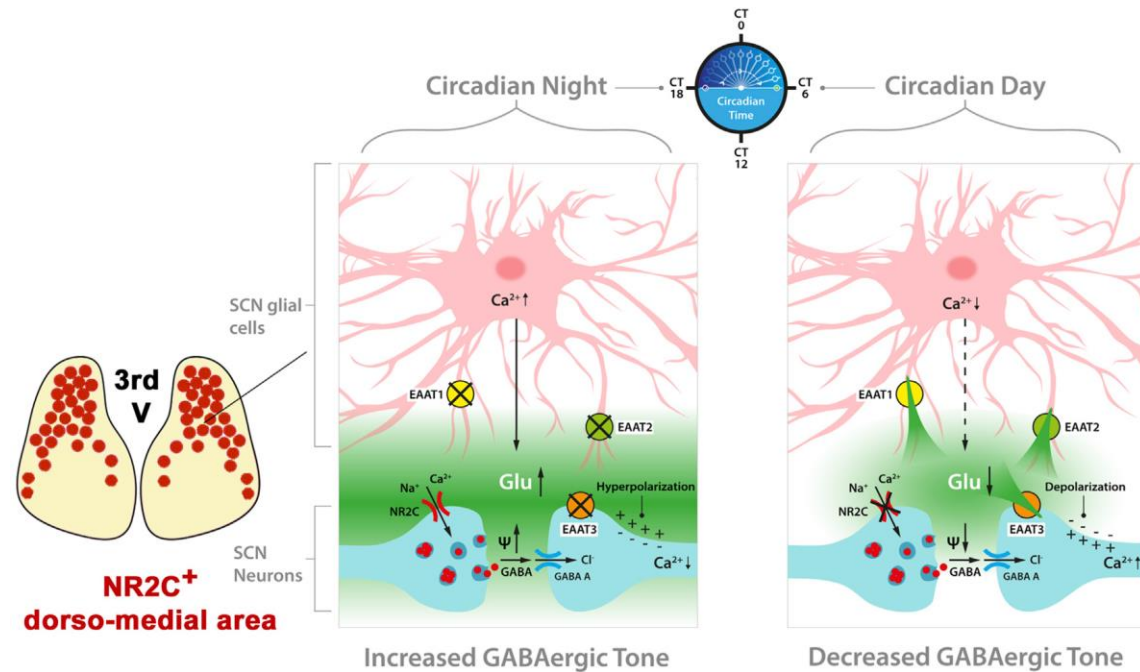


Major Hypothalamic Nuclei – Circadian rhythm

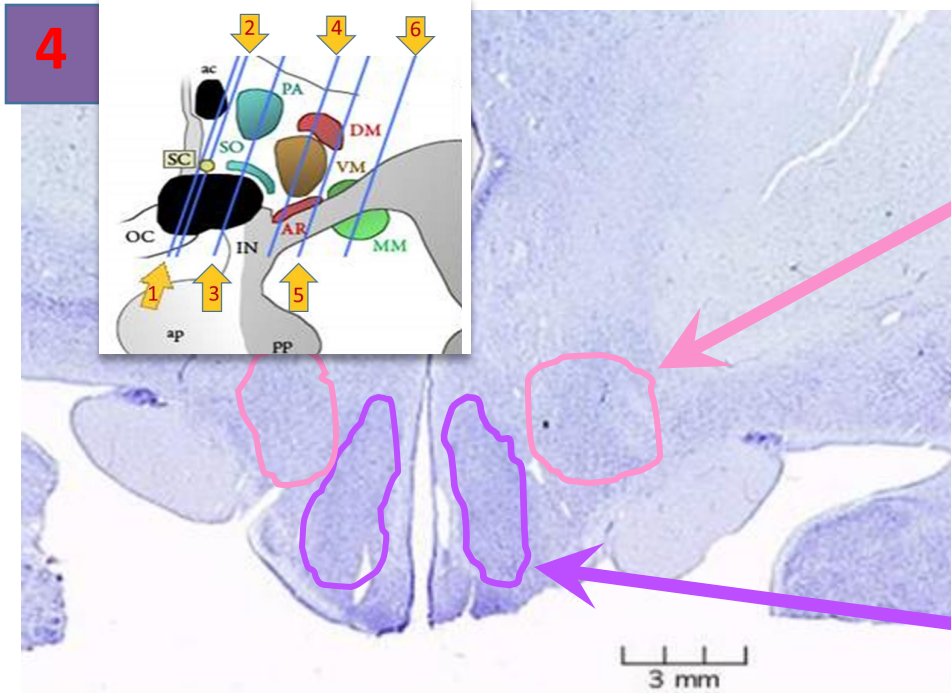
Suprachiasmatic Nucleus



- Receives direct input from the retina
- Plays a role in the regulation of circadian rhythms
- SCN astrocytes control circadian timekeeping



Major Hypothalamic Nuclei – Feeding & Arousal



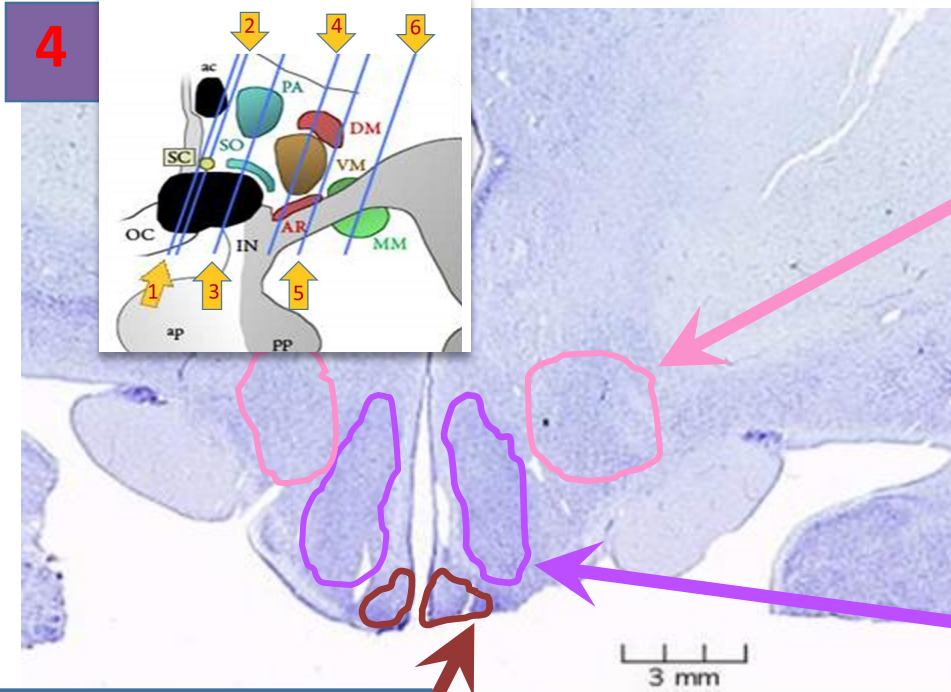
Lateral Hypothalamic Nucleus

- Feeding center of the brain
 - Lesions cause anorexia and weight loss
- Arousal center of the brain
 - Lesion/reduced number of neurons cause narcolepsy-like phenotype

Ventromedial Nucleus

- Satiety center of the brain
- Lesions cause hyperphagia (overeating) and obesity
- *Synthesizes somatostatin (SS)*

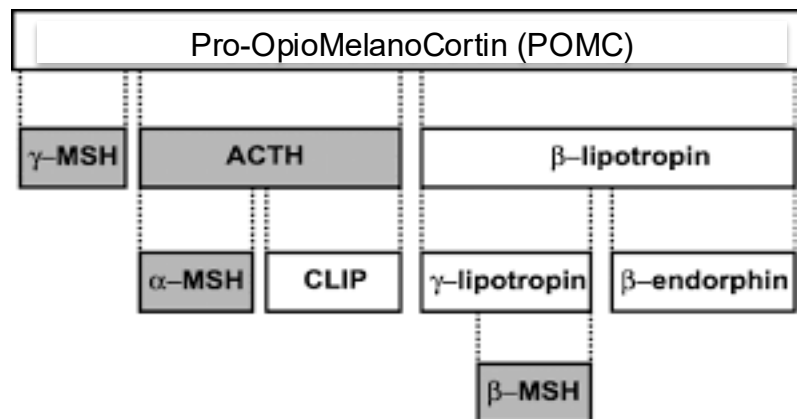
Major Hypothalamic Nuclei – Feeding & Arousal



Lateral Hypothalamic Nucleus

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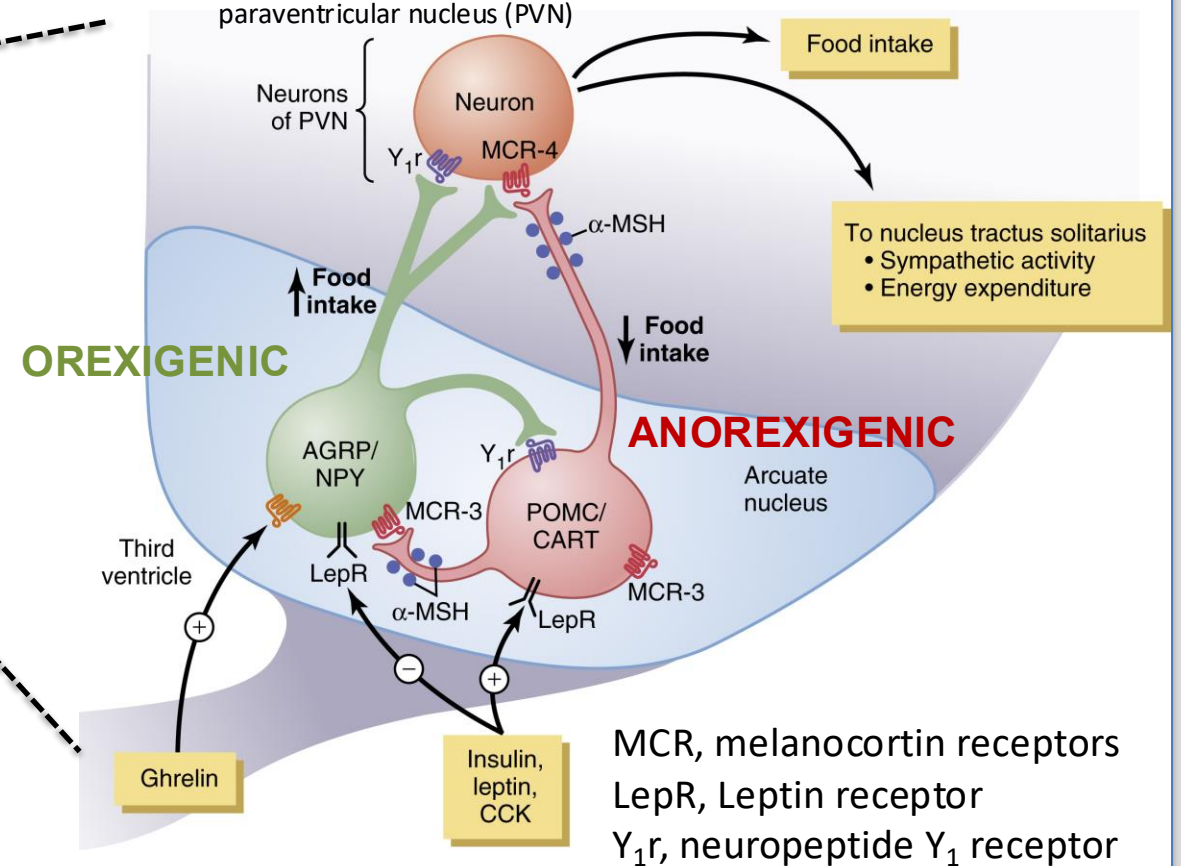
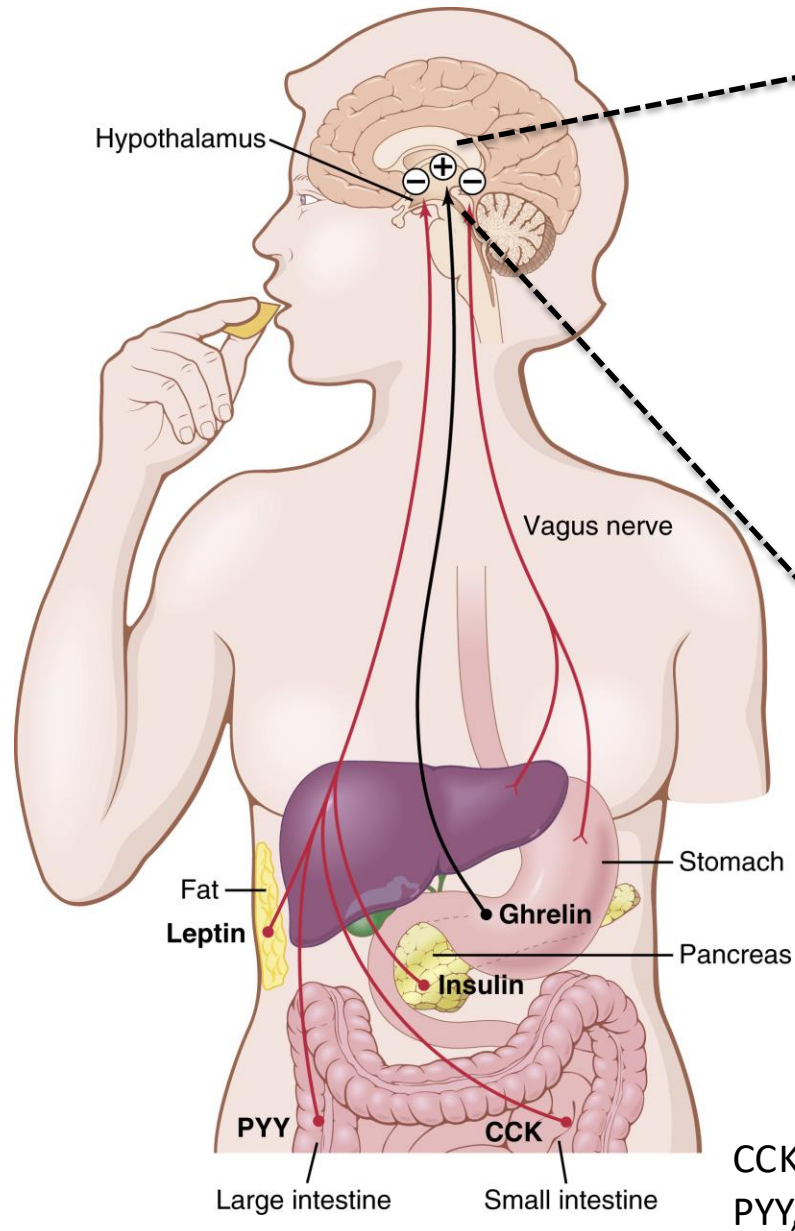
Arcuate Nucleus



Ventromedial Nucleus

- Satiety center of the brain
- Lesions cause hyperphagia (overeating) and obesity
- *Synthesizes somatostatin (SS)*

Energy balance and feeding



AGRP, agouti-related protein (MCR antagonist)
NPY, neuropeptide Y

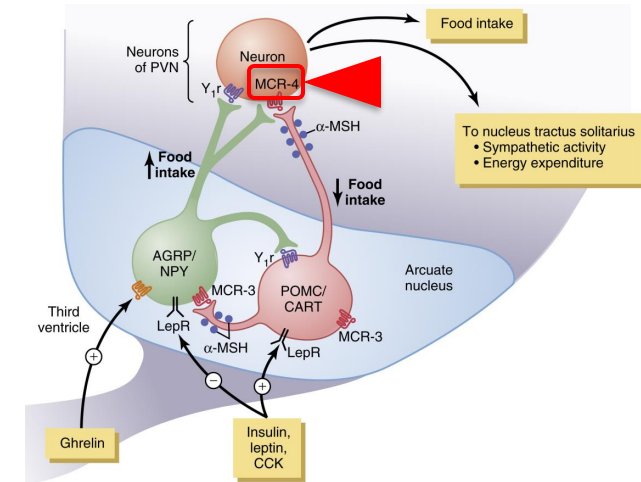
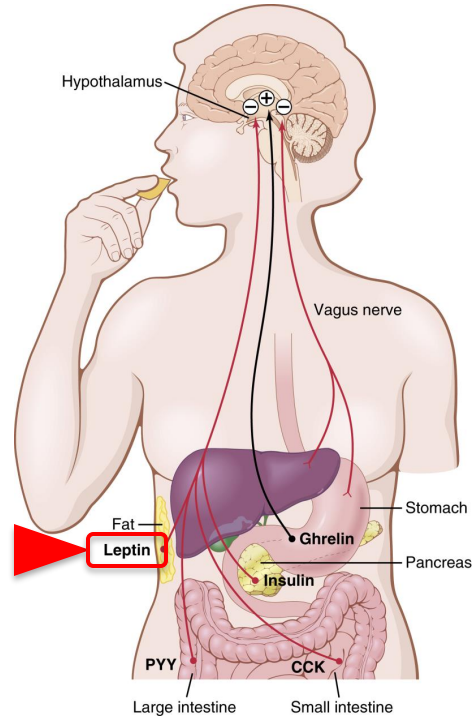
POMC, pro-opiomelanocortin

α-MSH, α-melanocyte-stimulating hormone

CART, cocaine- and amphetamine-regulated transcript

CCK, cholecystokinin
PYY, peptide YY

Energy balance and feeding – clinical relevance



MCR-4 Deficiency

* *most common form of monogenic obesity*

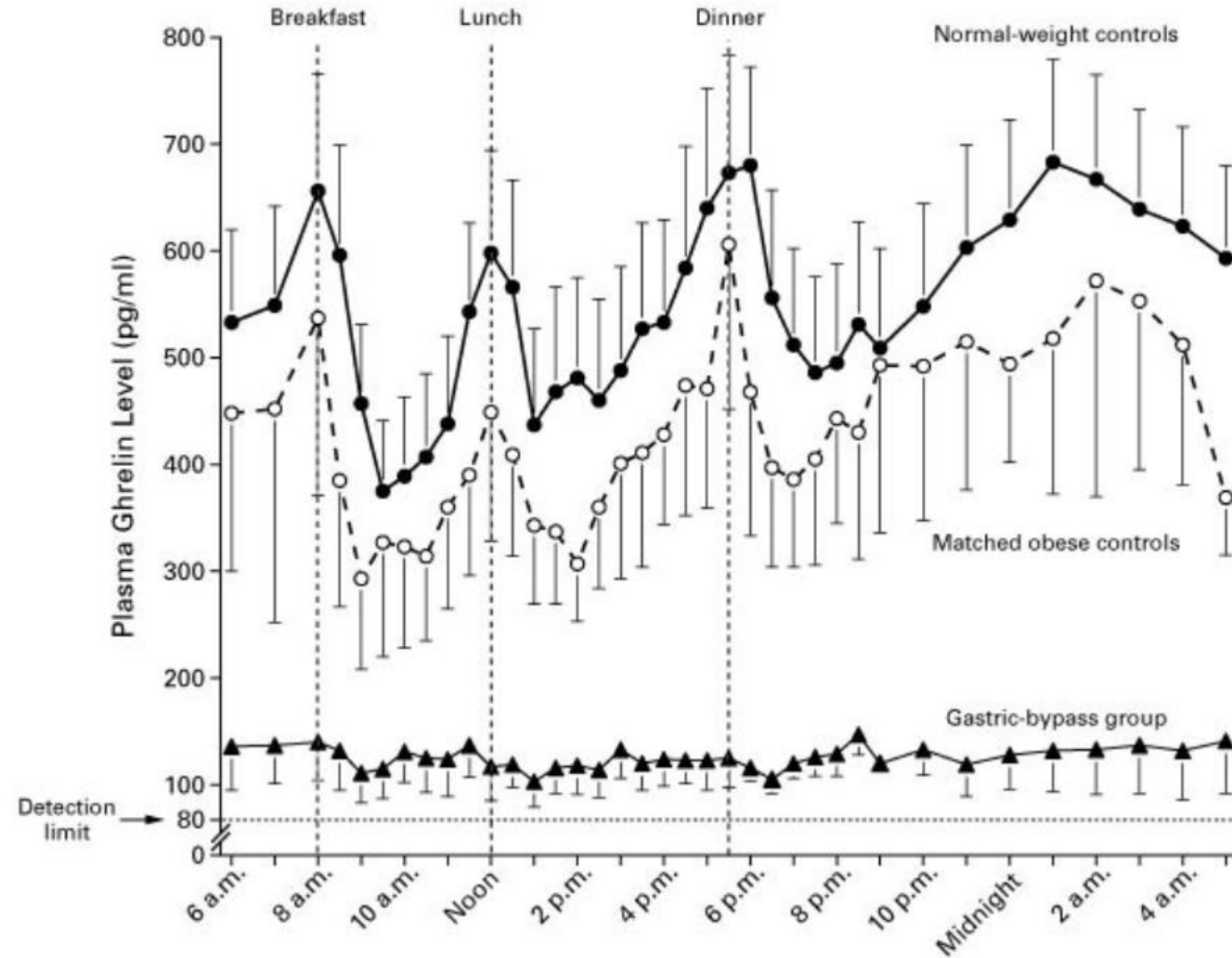
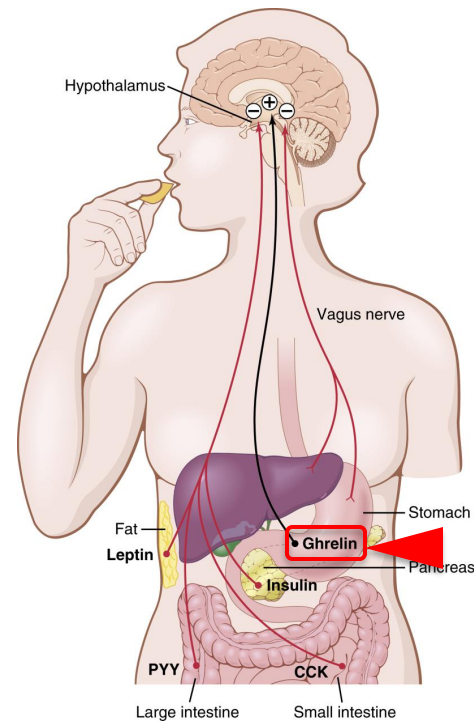
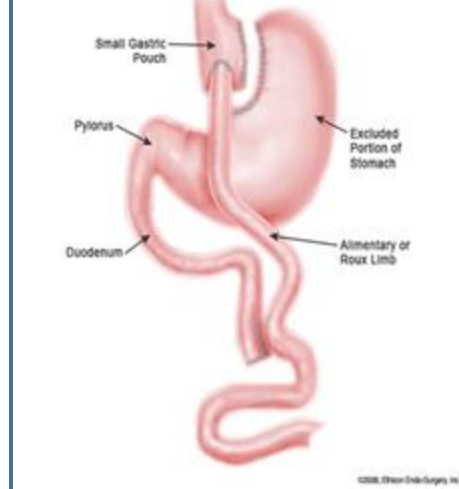
- Hyperphagia → obesity
- Hyperinsulinemia
- Accelerated linear growth in early childhood

Congenital Leptin Deficiency

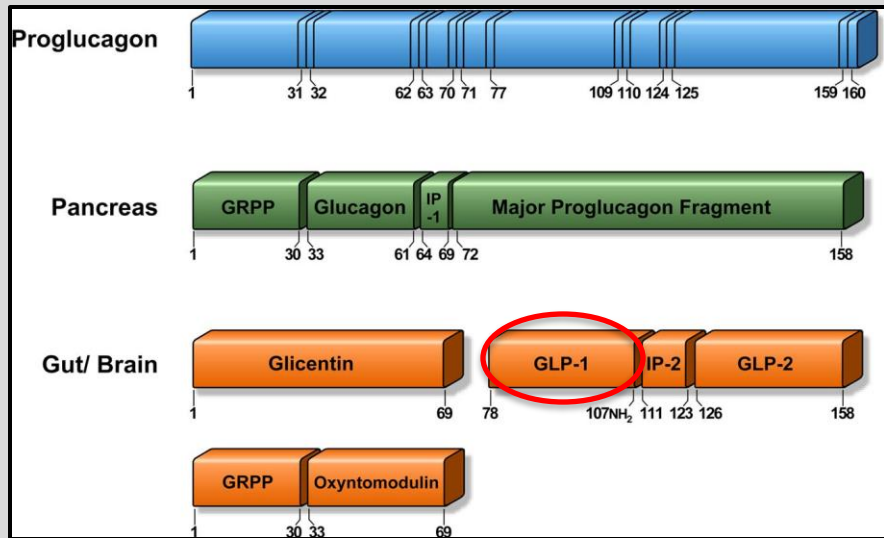
- Absence of **Leptin** → hyperphagia
- **Treatment:** subcutaneous recombinant human leptin

Energy balance and feeding – clinical relevance

Gastric Bypass Surgery

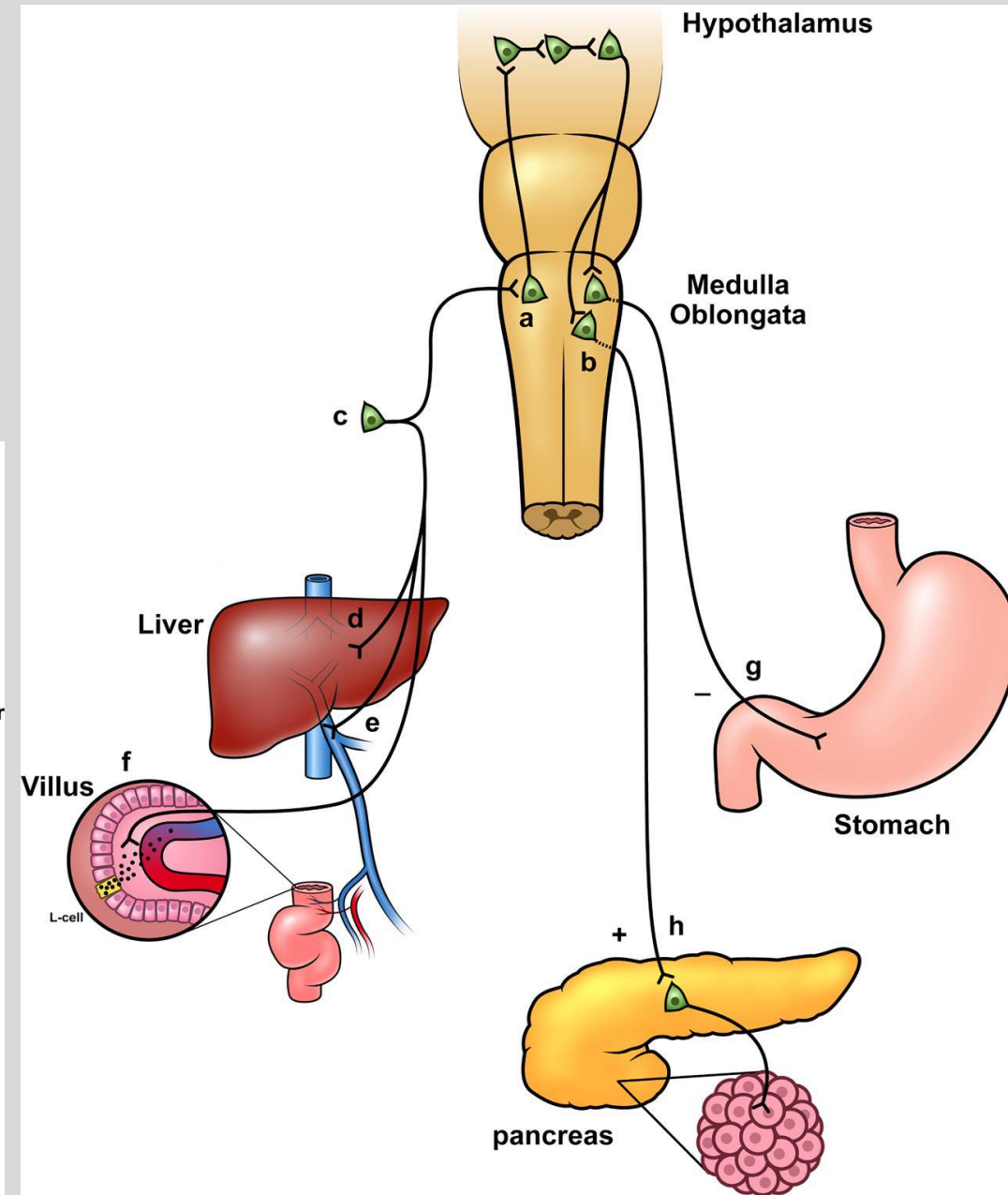
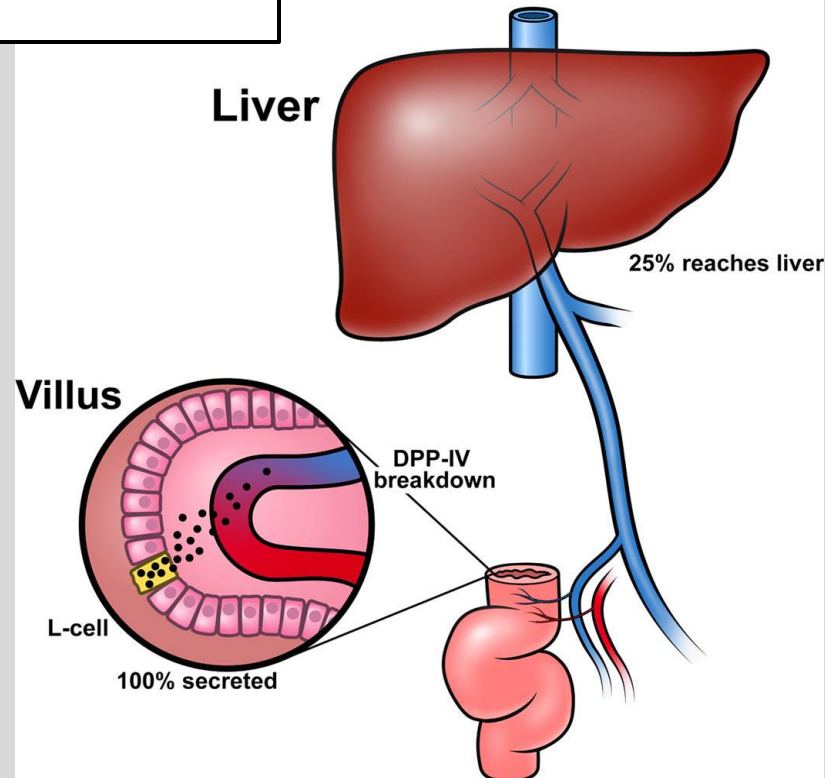


The Physiology of Glucagon-like Peptide 1 (GLP1)



Physiol Rev 87: 1409–1439, 2007.

10-15% of secreted amount leaves liver

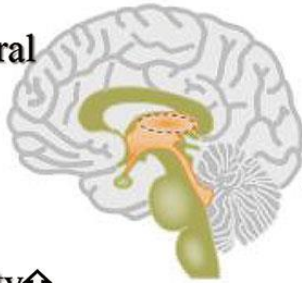


The effects of GLP1 and GLP1 receptor agonists

Exenatide
Liraglutide
Lixisenatide
Albiglutide
Dulaglutide
Semaglutide
Beinaglutide
Supaglutide
PEG-loxenatide

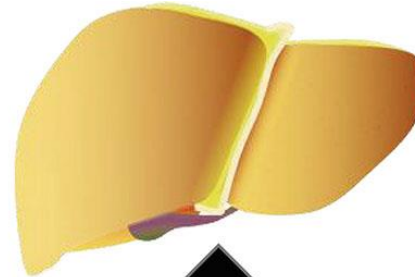
BRAIN

- Synaptic transmission ↑
- Nerve protection ↑
- Learning defects ↓
- Memory ↑
- The proliferation of neural stem cells ↑
- Memory impairment ↓
- Neuroinflammation ↓
- Motor function ↑
- The signal level of satiety ↑



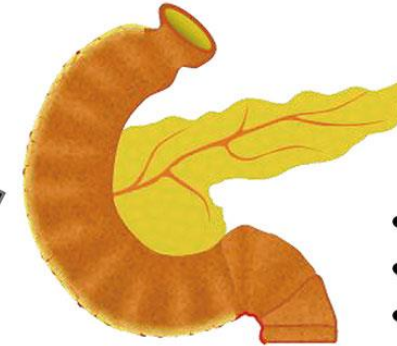
LIVER

- Liver glucose production ↓
- Liver fat content ↓
- Plasma liver enzyme level ↓
- Hepatic steatosis ↓



PANCREAS

- Insulin synthesis ↑
- Insulin secretion ↑
- Blood glucose ↓
- Islet β cell protection ↑
- Islet β cell proliferation ↑

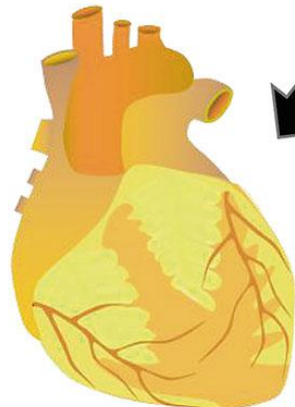


GLP-1



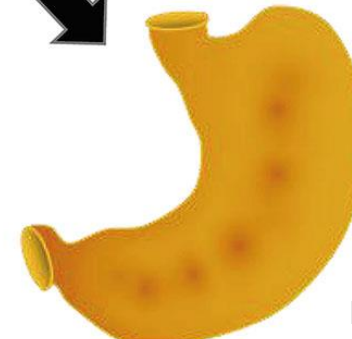
HEART

- Cardiovascular protection ↑
- Heart rate ↑
- Anti-inflammatory action ↑
- Myocardial ischemia injury ↓
- Endothelial dysfunction ↓
- Blood lipid ↓
- Atrial natriuretic peptide secretion ↑



GI TRACK

- Gastric emptying ↓
- Gastrointestinal peristalsis ↓



Tanycytes control hypothalamic liraglutide uptake and its anti-obesity actions

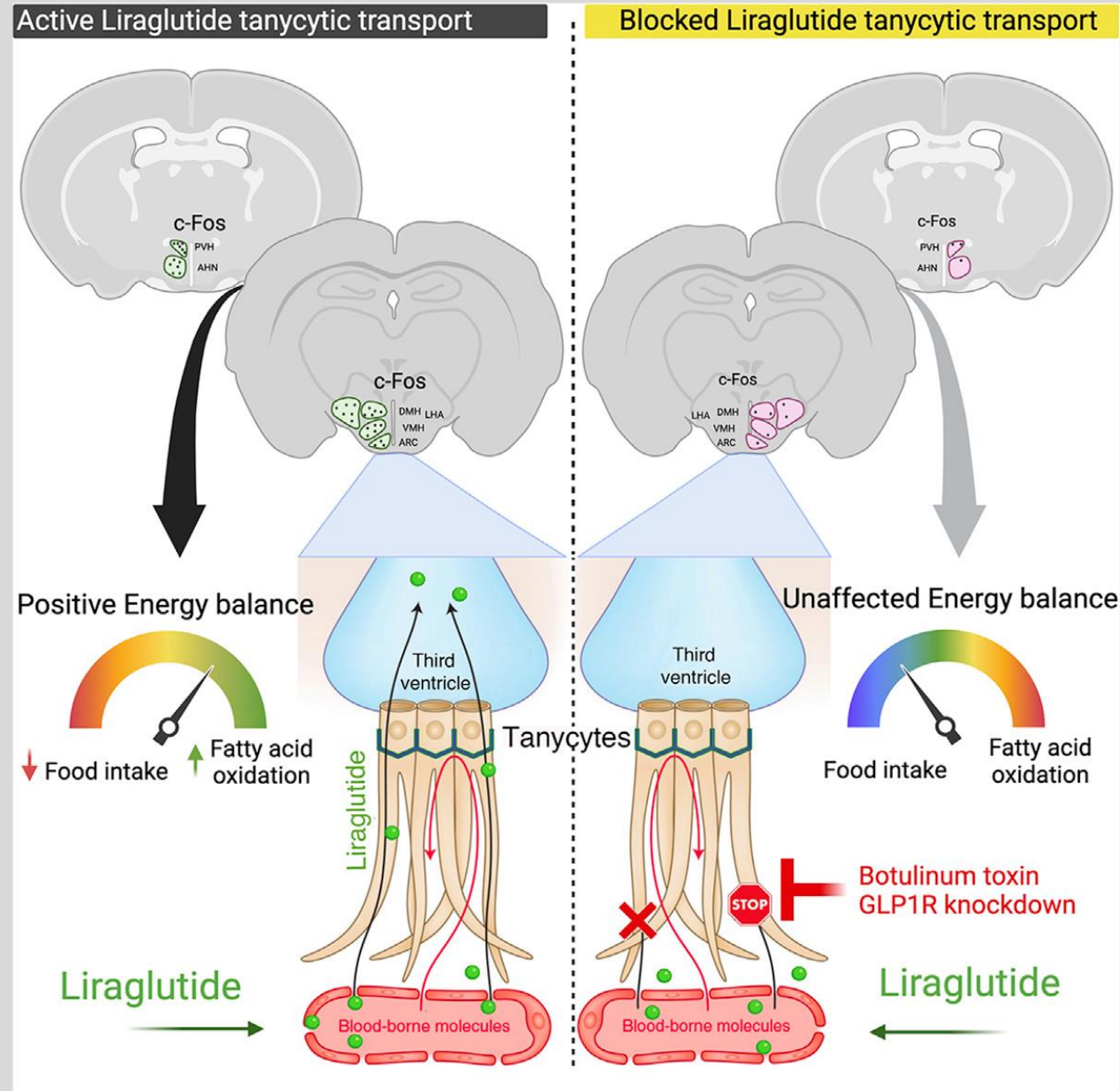


TABLE 7.6 Etiology of Hypothalamic Disease by Age

Premature Infants and Neonates

Intraventricular hemorrhage
Meningitis: bacterial
Tumors: glioma, hemangioma
Trauma
Hydrocephalus kernicterus

Age 1 Month to 2 Years

Tumors: optic glioma, histiocytosis X, hemangioma
Hydrocephalus
Meningitis
Congenital disorders: Kallmann syndrome, Laurence-Moon-Biedl syndrome, Prader-Willi syndrome

Age 2 to 10 Years

Tumors: craniopharyngioma, glioma, dysgerminoma, hamartoma, leukemia, histiocytosis X, ganglioneuroma, ependymoma, medulloblastoma
Meningitis: bacterial, tuberculous
Encephalitis: viral (exanthematous demyelinating)
Congenital diabetes insipidus
Radiation therapy
Diabetic ketoacidosis
Moyamoya disease, circle of Willis

Age 10 to 25 Years

Tumors: craniopharyngioma, glioma, hamartoma, dysgerminoma, histiocytosis X, leukemia, dermoid, lipoma, neuroblastoma
Trauma
Vascular: aneurysm, subarachnoid hemorrhage, arteriovenous malformation
Inflammatory disease: meningitis, encephalitis, sarcoidosis, tuberculosis
Structural brain defects: chronic hydrocephalus, increased intracranial pressure

Age 25 to 50 Years

Nutritional: Wernicke disease
Tumors: glioma, lymphoma, meningioma, craniopharyngioma, pituitary tumors, plasmacytoma, angioma, colloid cysts, sarcoma, ependymoma, histiocytosis X
Inflammatory diseases: sarcoidosis, tuberculosis, viral encephalitis
Vascular: aneurysm, subarachnoid hemorrhage, arteriovenous malformation
Damage from pituitary radiation therapy

Age 50 Years and Older

Nutritional: Wernicke disease
Tumors: pituitary tumors, sarcoma, glioblastoma, ependymoma, meningioma, colloid cysts, lymphoma
Vascular disease: infarct, subarachnoid hemorrhage, pituitary apoplexy
Inflammatory disease: encephalitis, sarcoidosis, meningitis
Damage from radiation therapy for ear-nose-throat carcinoma, pituitary tumors

Adapted from Plum F, Van Uiter R. Nonendocrine diseases and disorders of the hypothalamus. In: Reichlin S, Baldessarini RJ, Martin JB, eds. *The Hypothalamus*, vol 56. New York: Raven Press; 1978:415–473.

SUMMARY

Hypothalamic Nucleus/Area	Function
Anterior Hypothalamic Nucleus	Temperature regulation through cooling
Posterior Hypothalamic Nucleus	Temperature regulation through conservation and production of heat
Supraoptic Nucleus	AVP/ADH and oxytocin synthesis
Paraventricular Nucleus	AVP/ADH, CRF, TRH, and oxytocin synthesis; projections to autonomic nuclei
Medial Preoptic Nucleus	Regulation of gonadotropic hormone secretion
Arcuate Nucleus	Production of stimulatory or inhibitory hypothalamic factors, feeding
Dorsomedial Nucleus	Role in aggressive behavior, reproduction, feeding behavior, stress response
Suprachiasmatic Nucleus	Regulation of circadian rhythms
Mammillary Nuclei	Input from hippocampus, projections to thalamus; important in episodic memory
Ventromedial Nucleus	Satiety center, synthesizes somatostatin
Lateral Hypothalamic Nucleus	Feeding center, arousal center

Lecture Feedback Form:



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>